

Lec_12

0-1 $\Rightarrow$ Lecture

Lec_12

Omega and Theta notation

Big $\Rightarrow$ 0 ✓
small $\Rightarrow$ 0

Big $\Rightarrow$ Ω
small $\Rightarrow$ ω

if $\Rightarrow$ 0

$f(n) \leq c \cdot g(n)$
 $\Rightarrow$
 $f(n) = O(g(n))$

$n \geq n_0$
/)

2. Omega notation :

Let two non negative functions $f(n)$ and $g(n)$:

if $\Omega \Rightarrow \text{min}^m \text{ case}$

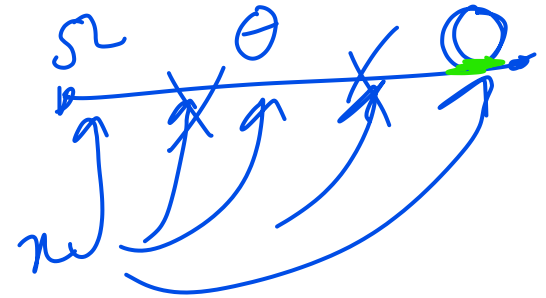
$$f(n) \geq c \cdot g(n)$$
$$\Downarrow$$
$$f(n) = \Omega(g(n))$$

n_0 is constant, $n_0 \neq \emptyset$

c is constant

$$c > 0$$

$$n \geq n_0$$



Graph

$$f(n) = n + 10, \quad g(n) = n + 30$$

$$f(n) \geq c \cdot g(n)$$

$$\Rightarrow n + 10 \geq c \cdot (n + 30)$$

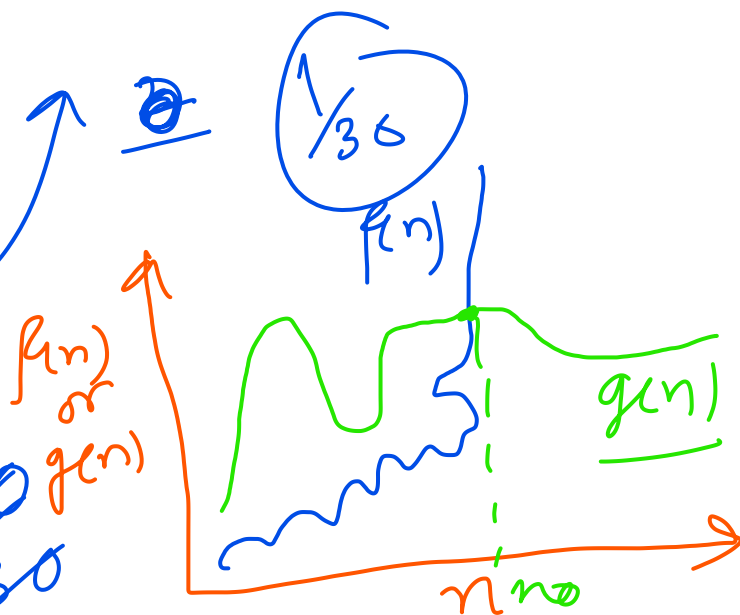
$$n + 10 \geq \frac{1}{30} (n + 30)$$

$$n + 10 \geq \frac{n}{30} + \frac{30}{30}$$

$$n + 10 \geq \frac{n}{30} + 1$$

$$n + 10 \geq c \cdot (n + 30)$$

$$f(n) = O(g(n))$$



$$f(n) \geq c \cdot g(n) \quad (n_0 \leq n)$$

$$f(n) = \Omega(g(n)) \quad n \geq n_0$$

Ex:1

$$F(n)=n+10, g(n)=n$$

$$\underline{f(n) = \Omega(g(n)) ?}$$

$$\underline{f(n) = O(g(n)) ?}$$

✓
vado
✓
last

$$f(n) \geq c \cdot g(n)$$
$$n+10 \geq c \cdot n$$

$$\Rightarrow \underline{c=1}$$

$$\underline{n+10 > n}$$
$$\underline{f(n) = \Omega(g(n))}$$

Ex:2

$$F(n)=n^2+n+10, g(n)=n^2$$

$$\begin{aligned} f(n) &\geq c \cdot g(n) & c=1 \\ n^2+n+10 &\geq c \cdot n^2 \\ \hline n^2+n+10 &\geq n^2 \end{aligned}$$

$$\begin{aligned} f(n) &= \Omega(g(n)) \\ f(n) &\geq c \cdot g(n) \end{aligned}$$

Ex-3: $f(n)=n^2$, $g(n)=n$

~~find~~

$$\underline{f(n) = \Omega(g(n))}$$

$$f(n) = n^2, g(n) = n$$

$$\frac{f(n) \geq c \cdot g(n)}{n^2 \geq c \cdot n} \quad c=1$$

Ex:34

$$F(n)=n-10, g(n)=n+10$$

$$\begin{aligned} n-10 &\geq (n+10) \\ \Rightarrow n-10 &\geq \frac{(n+10)}{10} \quad \text{no?} \\ n-10 &\geq \frac{n}{10} + 1 \quad c=1/10 \end{aligned}$$

$\Rightarrow$ if $\Rightarrow f(n) \geq c \cdot g(n)$
 $\Downarrow$
 $f(n) = \Omega(g(n))$

$\Leftarrow$ if $f(n) \leq c \cdot g(n)$
 $\Downarrow$
 $f(n) = O(g(n))$

2. Theta notation : Θ

Let two non negative functions $f(n)$ and $g(n)$:

$\Theta \Rightarrow$ Avg Case

if $f(n)$, $g(n)$... $\rightarrow C1$

$f(n) \geq c_1 g(n)$ $\rightarrow C2$

and

$f(n) \leq c_2 g(n)$

then

$f(n) = \Theta(g(n))$

$g(n) = \Theta(f(n)) \Rightarrow$ Both are true.

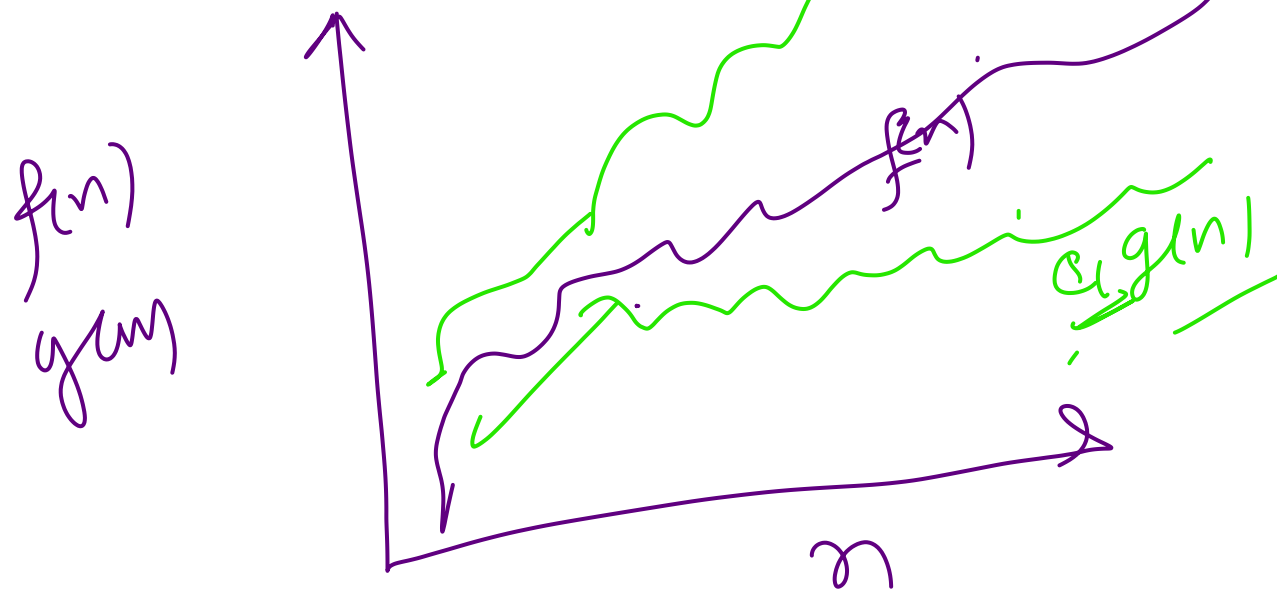
Because of them.

$f(n) = n^2$
 $g(n) = 2n^2$

$f(n) \leq c_2 g(n)$
 $n^2 \leq 1/2 n^2$

$f(n) \geq c_1 g(n)$
 $n^2 \geq 2n^2$
 $n^2 \geq \frac{1}{2} n^2$

Graph



⊕ condition

$f(n) \geq c_1 g(n)$ and $f(n) \leq c_2 g(n)$
you can choose

Ex: $f(n)=n, g(n)=n$

cond¹: $f(n) \geq c_1 g(n)$
 $n \geq \frac{c_1 n}{2} \Rightarrow c_1 = \frac{1}{2}$
 $n \geq \frac{n}{2}$

cond²: $f(n) \leq c_2 g(n)$
 $n \leq c_2 g(n) \rightarrow c_2 = 2$
 $n \leq 2n$

we can say $\Rightarrow$

$$\boxed{\begin{array}{l} f(n) = \theta(g(n)) \\ \Downarrow \\ g(n) = \theta(f(n)) \end{array}}$$

Ex.2: $f(n)=n^2+2$, $g(n)=n^2$

c_1 :

$$f(n) \geq c_1 g(n)$$
$$n^2 + 2 \geq c_1 n^2$$

$$\Rightarrow c_1 = 1$$

$$n^2 + 2 \geq n^2$$
$$f(n) \leq c_2 g(n)$$

c_2 :

$$n^2 + 2 \leq 2n^2$$

$$c_2 = 2$$

$$f(n) = \Theta(g(n))$$
$$g(n) = \Theta(f(n))$$

$f(n)$ and $g(n)$ $\bigcirc$ ~~****~~

if $f(n) \leq c g(n)$

$$f(n) = O(g(n))$$

$c > 0$
 $n \geq n_0$
 $n_0 \geq 1$

QED

$\therefore \Sigma$
if $f(n) \geq c g(n)$
 $f(n) = \Omega(g(n))$

$c = \text{const}$

$c > 0$

$n_0 \geq 1$

$\oplus$
if $f(n) \geq c_1 g(n)$
and
 $f(n) \leq c_2 g(n)$
 $\Downarrow$
 $f(n) = \Theta(g(n))$
or
 $g(n) = \Theta(f(n))$

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